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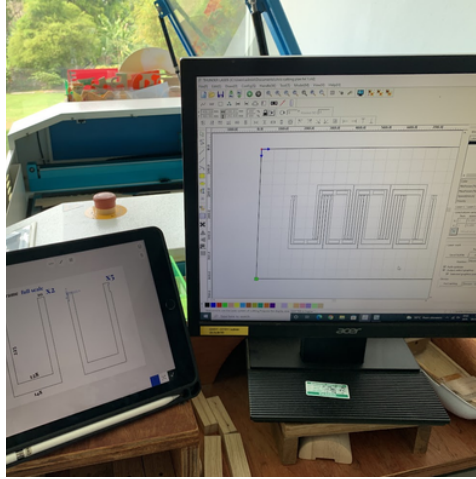
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Folding Lamp

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IGCSE Design & Technology Final Product

St Andrews International School Green Valley



What?

- Identify a real-world problem.
- Create a design folder containing research and annotated concept sketches.
- Build a physical **prototype**.
- **Evaluate** your finished outcome against a strict specification.

How?

- Sketched and designed 8 concepts with **Fusion 360**
- Designed and prototyped a folding lamp using 3D-printed, stacked acrylic, and wooden parts; custom hidden hinges for full-range support.
- Optimised laser-cut layout to cut material waste by 50%.
- Resoldered a solar lamp, battery, button and PCB board to remove the motion sensor.

Results

- Produced a versatile 360-degree desk lamp that can collapse into a backpack.
- Custom warm light colour filter to reduce eye strain.
- Easy to use with minimal pinch points to improve safety.